

A stability refinement for simple critical zeros in short intervals

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Abstract

For each fixed θ for which Wang’s short-interval cosine bound $c(\theta) = 2 - \theta/2 - \cot(\theta/\sqrt{2})/\sqrt{2}$ is positive, we give an explicit constant strictly larger than $c(\theta)$ as a lower asymptotic proportion of simple critical zeros in $(T, T + T^\theta]$. The corresponding distinct-zero bound also improves. We transfer the known stable rank–trace inequality in the **ainta** manuscript directly to Lamzouri’s finite Hilbert operator, retain the Gram defect of its simple real points, and combine three shifted partitions into consecutive triples. A quantitative obstruction to additive triples of cosine-kernel zeros supplies a positive gap for every fixed admissible θ . Wang’s fixed-test pair-correlation theorem provides the arithmetic input; smoothing is performed after the large-height limit. An interval certificate at $\theta = 3/4$ improves the simple-critical constant from $0.4190750129754243\dots$ to $0.4190768284253039\dots$, and gives distinct proportion $0.7095384142126519\dots$. The general strict improvement has an analytic proof independent of numerical minimization. This does not improve the positivity exponent, claim a global record, or resolve the Riemann hypothesis.

1 Statement and relation to prior work

Let $N(T, H)$ count the nontrivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $T < \gamma \leq T + H$, including multiplicity. Let $N_0^s(T, H)$ count the simple zeros among them with $\beta = 1/2$, and let $N^d(T, H)$ count distinct complex zeros, each once. Every proportion below has denominator $N(T, H)$.

The recent argument discovered by Claude and communicated by Alpöge and Furman [1] made the optimized Montgomery–Taylor simple-critical proportion $0.672500703679\dots$ unconditional. Lamzouri [2] replaced its finite-compression machinery by a Hilbert-space inequality for conjugation-invariant multisets. Wang [3] established the needed short-interval pair-correlation theorem and deduced, for fixed $0 < \theta < 1$,

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq c(\theta), \quad \liminf_{T \rightarrow \infty} \frac{N^d(T, T^\theta)}{N(T, T^\theta)} \geq \frac{1 + c(\theta)}{2}, \quad (1)$$

where

$$c(\theta) = 2 - \frac{\theta}{2} - \frac{1}{\sqrt{2}} \cot(\theta/\sqrt{2}). \quad (2)$$

The derivative is $c'(\theta) = \frac{1}{2} \cot^2(\theta/\sqrt{2}) > 0$. Since $c(\theta) \rightarrow -\infty$ as $\theta \downarrow 0$ and $c(1) > 0$, there is a unique zero $\theta_0 \in (0, 1)$, numerically $\theta_0 = 0.550193964744154\dots$

The stable rank–trace inequality, convex Gram defect, pinching method, and cosine-root non-additivity used here are prior work in the `ainta` repository [4]. Further global window and block refinements are also under active development [5]. We do not claim these ingredients or an improved global record. Our contribution is their direct finite-Hilbert transfer, an explicit quantitative three-point bound, and the strict refinement of Wang’s entire positive short-interval curve. The finite proof below is included to make the treatment of off-line zeros, multiplicities, and singular Gram matrices explicit.

Theorem 1.1 (Strict short-interval refinement, D). *Fix $\theta_0 < \theta < 1$ and $R > 2/c(\theta)$. Define*

$$\begin{aligned} a &= \theta/\sqrt{2}, & q &= a \tan a, & M &= \pi\theta R, \\ B &= q(1 + M^2/a^2), \\ b &= \min \left\{ \frac{q}{2}, \frac{q^4}{4(q+M)^2(M+3q/2)} \right\}, & \delta &= \frac{1}{2} \min\{1, 2(b/B)^2\}. \end{aligned} \quad (3)$$

Then $0 < \delta \leq 1/2$, and, with

$$c_*(\theta, R) = c(\theta) + \frac{\delta\{c(\theta) - 2/R\}}{3 - \delta} > c(\theta), \quad (4)$$

we have

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq c_*(\theta, R), \quad \liminf_{T \rightarrow \infty} \frac{N^d(T, T^\theta)}{N(T, T^\theta)} \geq \frac{1 + c_*(\theta, R)}{2}. \quad (5)$$

In particular, $R = 4/c(\theta)$ gives a fully specified positive improvement.

The labels [D] and [MV] distinguish mathematical deductions from finite machine-verifiable statements. The conservative gap in Theorem 1.1 requires no numerical search. A stronger finite certificate gives the following example.

Corollary 1.2 (Certified example, D+MV). *Set $\theta = 3/4$, $R = 21/4$, $d = 1/7000$, and*

$$C_* := \frac{3c(3/4) - 2d/R}{3 - d} = 0.419076828425303996736665787527\dots$$

Then the simple-critical proportion in (5) is at least C_* , and the distinct-zero proportion is at least

$$\frac{1 + C_*}{2} = 0.709538414212651998368332893763\dots$$

The improvement in the simple-critical proportion is $0.000001815449879663002112176828\dots$

The conclusion is for each fixed exponent and all sufficiently large heights in the asymptotic sense. We do not provide an effective starting height or uniformity as $\theta \downarrow \theta_0$. In particular, this does not lower Wang’s positive-proportion threshold. The analytic dependency is Wang’s inspected preprint theorem, whose underlying pair-correlation method is due to Baluyot, Goldston, Suriajaya, and Turnage-Butterbaugh [6, 7].

2 A stable inequality for finite multisets

Let Z be a finite conjugation-invariant multiset in $\mathbb{C}$. We sum over its distinct support with positive integer multiplicities $m_z = m_{\bar{z}}$, and put $N = \sum_z m_z$. Choose a real even compactly supported $\eta \in L^2(\mathbb{R})$ with $\int \eta^2 = 1$, and define

$$v_z(u) = \eta(u)e^{-2\pi izu}, \quad K(t) = \widehat{\eta^2}(t), \quad \widehat{f}(t) = \int_{\mathbb{R}} f(u)e^{-2\pi itu} du.$$

Every v_z belongs to L^2 because Z is finite and the support is compact. Work in the finite-dimensional span E of these vectors. Our inner product is $\langle v, w \rangle = \int \bar{v}w$, linear in the second variable; write $|v\rangle\langle w|$ for $h \mapsto v\langle w, h \rangle$.

Define the reflected rank-one operator

$$A = \sum_z m_z |v_z\rangle\langle v_{\bar{z}}|. \quad (6)$$

It is selfadjoint by conjugation invariance. Direct multiplication gives

$$\mathrm{tr} A = N, \quad Q := \|A\|_{\mathrm{HS}}^2 = \mathrm{tr} A^2 = \sum_{z,w} m_z m_w K(z-w)^2. \quad (7)$$

Indeed, $\langle v_{\bar{z}}, v_z \rangle = \int \eta^2 = 1$, while the trace of the product of the z and w terms, before multiplicity factors, is $K(w-z)K(z-w) = K(z-w)^2$, since K is even. The ordinary square in (7) is essential. Individual terms for complex arguments need not be positive or even real; their complete sum is a nonnegative operator norm.

Let $x_1 < \dots < x_s$ be the simple real points of Z . Let r be the number of distinct multiple real points and k the number of distinct nonreal conjugate pairs. Put

$$P = \sum_{j=1}^s |v_{x_j}\rangle\langle v_{x_j}| = VV^*, \quad G = V^*V = (K(x_i - x_j))_{i,j=1}^s, \quad C = A - P.$$

Then G is positive semidefinite with diagonal one and $\mathrm{tr} P = s$. For each nonreal pair put $g = (v_z + v_{\bar{z}})/2$ and $h = (v_z - v_{\bar{z}})/(2i)$. Its contribution to C is $2m_z(|g\rangle\langle g| - |h\rangle\langle h|)$. The form of C is nonpositive on the orthogonal complement of the r multiple-real vectors and k vectors g . Thus

$$n_+(C) \leq r + k, \quad N \geq s + 2r + 2k. \quad (8)$$

No distance of a nonreal point from the real axis is assumed.

Lemma 2.1 (Known stable rank–trace inequality). *Suppose $P = VV^*$ on a finite-dimensional Hilbert space, the s columns of V have norm one, C is selfadjoint, and $n_+(C) \leq d$. Define*

$$\Psi(t) = \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2. \end{cases}$$

Then

$$\|P + C\|_{\mathrm{HS}}^2 \geq 4\mathrm{tr}(P + C) - 3s - 4d + \mathrm{tr} \Psi(V^*V). \quad (9)$$

Proof. This is the unit-column case of the stability inequality in [4, Section 2]. We give a spectral proof that keeps all zero eigenvalues. Let $m = \dim E$ and let $p_1 \leq \dots \leq p_m$ be the eigenvalues of

P . If $d < m$, the min-max principle gives $\lambda_i(P + C) \leq p_{i+d}$ for $1 \leq i \leq m - d$: intersect the span of the first $i + d$ eigenvectors of P with a codimension-at-most- d subspace on which $C \leq 0$.

It follows that

$$\operatorname{tr}(P + C - 2I)^2 \geq \sum_{j=d+1}^m (2 - p_j)_+^2 \geq \sum_{j=1}^m (2 - p_j)_+^2 - 4d.$$

Here $0 \leq (2 - p_j)_+^2 \leq 4$. If $d \geq m$, the final right-hand side is nonpositive and the same inequality is immediate. For every $p \geq 0$, $(2 - p)_+^2 = \Psi(p) - 2p + 3$. The nonzero spectra of VV^* and V^*V agree, and $\Psi(0) = 1$, so

$$\operatorname{tr} \Psi(P) = \operatorname{tr} \Psi(V^*V) + m - s.$$

Substitute these identities and $\operatorname{tr} P = s$, and expand the left-hand square. All dimensional terms cancel and give (9). $\square$

Proposition 2.2 (Finite Hilbert stability transfer). *For the multiset and operator above, write $D(G) = \operatorname{tr} \Psi(G)$ and $D_Z = s + r + 2k$ for the number of distinct support points. Then*

$$s \geq 2N - Q + D(G), \quad D_Z \geq \frac{3N - Q + D(G)}{2}. \quad (10)$$

Proof. By Lemma 2.1 and (8), $Q \geq 4N - 3s - 4(r + k) + D(G)$. The multiplicity bound in (8) makes this at least $2N - s + D(G)$. For the second claim, the difference between $4N - 3s - 4(r + k)$ and $3N - 2D_Z$ is $N - s - 2r \geq 0$. $\square$

This proof retains the signed off-line remainder throughout. The positive Gram defect belongs only to the actual simple real points. The second inequality in (10), rather than elementary counting from a simple-zero proportion, will yield the distinct-zero companion.

3 From three-point energy to a density gain

For a positive semidefinite matrix M with diagonal one, define $\mathcal{E}(M) = 2 \sum_{i < j} |M_{ij}|^2$.

Lemma 3.1 (Block defect and pinching). *We have $D(M) \geq \min\{1, \mathcal{E}(M)\}$. If M_j are the principal blocks of a partition of M , then $D(M) \geq \sum_j D(M_j)$.*

Proof. If all eigenvalues of M are at most 2, then $D(M) = \operatorname{tr}(M - I)^2 = \mathcal{E}(M)$. If some eigenvalue exceeds 2, its own Ψ value exceeds 1 and all other terms are nonnegative. For pinching, diagonalize each principal block and let (w_i) be the resulting orthonormal basis of the full space. Scalar Jensen applied to the spectral resolution of M gives

$$\sum_j D(M_j) = \sum_i \Psi(\langle w_i, M w_i \rangle) \leq \sum_i \langle w_i, \Psi(M) w_i \rangle = D(M).$$

Only scalar convexity of Ψ is used; no operator-convex assertion is needed. $\square$

Lemma 3.2 (Three shifted partitions). *Suppose $0 < d \leq 1$, $R > 0$, and the real even kernel satisfies*

$$2\{K(u)^2 + K(v)^2 + K(u + v)^2\} \geq d, \quad u, v \geq 0, \quad u + v \leq R. \quad (11)$$

If $\ell = x_s - x_1$ is the span of the simple real points, with $\ell = 0$ when $s \leq 1$, then

$$D(G) \geq \frac{d}{3} \left(s - 2 - \frac{2\ell}{R} \right). \quad (12)$$

Proof. For $s \geq 3$, the sum of the spans of its $s - 2$ consecutive triples is at most 2ℓ , since each successive gap occurs at most twice. Hence at most $2\ell/R$ triples have span greater than R . Every remaining triple has defect at least d by Lemma 3.1.

Separate the starting indices of consecutive triples into their three residue classes modulo 3. Within each class the triples are disjoint principal blocks; add the unused points as singleton blocks. Apply pinching to each partition and average the three inequalities. Every short triple has been counted once, proving (12). If its right-hand side is negative, nonnegativity proves the inequality; this also covers $s < 3$. $\square$

The factor $1/3$ records the three shifted partitions. Overlapping triples cannot simply be summed without this step.

4 A quantitative obstruction for the cosine kernel

For $0 < \lambda \leq 1$ put

$$f_\lambda(u) = \frac{\cos(\sqrt{2}u)}{\sqrt{2}\sin(\lambda/\sqrt{2})} \mathbf{1}_{[-\lambda/2, \lambda/2]}(u), \quad K_\lambda = \widehat{f}_\lambda. \quad (13)$$

This is a nonnegative probability density. Set $a = \lambda/\sqrt{2}$, $q = a \tan a$, and $X = \pi\lambda t$. Integration yields

$$K_\lambda(t) = \frac{\cos X - (X/a) \cot(a) \sin X}{1 - X^2/a^2}. \quad (14)$$

The apparent singularities at $X = \pm a$ are removable. Equivalently, with $\text{sinc}(x) = \sin(x)/x$ continuously extended at zero,

$$K_\lambda(t) = \frac{\text{sinc}(X - a) + \text{sinc}(X + a)}{2 \text{sinc}(a)}. \quad (15)$$

An actual positive root satisfies $X \tan X = q$. If $X, Y, X + Y$ all corresponded to positive roots, tangent addition would force $X^2 + XY + Y^2 + q^2 = 0$. The following estimate gives a quantitative version without dividing by a possibly vanishing cosine or removable denominator.

Lemma 4.1 (Explicit positive triangle energy). *Fix $R > 0$, put $M = \pi\lambda R$, and define B, b as in (3), using $a = \lambda/\sqrt{2}$ and $q = a \tan a$. Then, throughout $u, v \geq 0$, $u + v \leq R$,*

$$2\{K_\lambda(u)^2 + K_\lambda(v)^2 + K_\lambda(u + v)^2\} \geq 2(b/B)^2 > 0. \quad (16)$$

Proof. Let $A_X = q \cos X - X \sin X$. For $Z = X + Y$, angle addition gives

$$(X^2 + XY + Y^2 + q^2) \sin X \sin Y = A_X A_Y - X A_X \sin Y - Y A_Y \sin X - q A_Z. \quad (17)$$

Take $X, Y \geq 0$, $Z \leq M$, and put $h = \max\{|A_X|, |A_Y|, |A_Z|\}$. If $h \leq q/2$, then

$$q \leq q|\cos X| + q|\sin X| \leq h + (q + X)|\sin X|,$$

so $|\sin X| \geq q/[2(q + M)]$, and likewise for Y . Taking absolute values in (17) therefore yields

$$\frac{q^4}{4(q + M)^2} \leq h^2 + (M + q)h \leq (M + 3q/2)h.$$

If $h > q/2$, the other alternative in the definition of b applies. In either case $h \geq b > 0$. The identity

$$A_X = q(1 - X^2/a^2)K_\lambda(X/(\pi\lambda))$$

holds also at the removable points by continuity. Since $|q(1 - X^2/a^2)| \leq B$ for $0 \leq X \leq M$, at least one of the three kernel values has absolute value at least b/B . This proves (16). $\square$

The zero-set obstruction itself is already used in [4]. The explicit formula in Lemma 4.1 makes its positive compact gap available for every fixed exponent without a numerical minimum. Its small size is acceptable for the qualitative strict-improvement theorem; Section 6 certifies a more useful gap in one example.

5 Short-interval arithmetic and passage to the sharp density

We record the exact analytic interface to avoid a support or smoothing uniformity assumption. Fix $0 < \lambda < \theta < 1$, put $H = T^\theta$, $L = \log T$, and $I = (T, T + H]$. Wang's Theorem 2.2 states that for every fixed real even $g \in C_c^\infty(\mathbb{R})$ supported in $[-\lambda, \lambda]$,

$$\begin{aligned} W_I(g) &:= \sum_{\gamma, \gamma' \in I} \widehat{g}\left(\frac{i(\rho - \rho')L}{2\pi}\right) \frac{4}{4 - (\rho - \rho')^2} \\ &= \frac{HL}{2\pi} \left(g(0) + \int_{\mathbb{R}} |\alpha| g(\alpha) d\alpha \right) + O_g(H + T^\lambda L^2). \end{aligned} \quad (18)$$

The constants may depend on fixed λ, θ . This is an unconditional complex-zero formula; RH is not an assumption.

Choose a fixed real even $\eta \in C_c^\infty((-\lambda/2, \lambda/2))$ with $\int \eta^2 = 1$, and write $f = \eta^2$, $K = \widehat{f}$, $g = f * f$. Both g and g'' meet the fixed-test hypotheses. At $z = i(\rho - \rho')L/(2\pi)$, differentiation of the Fourier transform gives

$$\widehat{g''}(z) = -4\pi^2 z^2 K(z)^2 = L^2(\rho - \rho')^2 K(z)^2.$$

The exact identity

$$\left(\widehat{g}(z) - \frac{\widehat{g''}(z)}{4L^2} \right) \frac{4}{4 - (\rho - \rho')^2} = K(z)^2$$

removes the rational weight. Apply (18) separately to the two fixed tests and combine the resulting estimates. As in [3, Lemma 3.1], this gives

$$\sum_{\gamma, \gamma' \in I} K(z_\rho - z_{\rho'})^2 = \mathcal{C}(f) \frac{HL}{2\pi} + O_f(H + T^\lambda L^2), \quad (19)$$

where

$$z_\rho = \frac{i(\rho - 1/2)L}{2\pi}, \quad \mathcal{C}(f) = \int f^2 + \iint |u - v| f(u) f(v) du dv. \quad (20)$$

Indeed, the main term for g'' is $-\int (f')^2 + 2\int f^2$, finite for the fixed smooth density, and its coefficient is $1/(4L^2)$. A varying scalar coefficient after applying the theorem to two fixed tests does not require a varying-test theorem.

Reflection $\rho \mapsto 1 - \bar{\rho}$ preserves the height interval and multiplicity and maps z_ρ to $\bar{z}_\rho$. Thus the multiset $Z_T = \{z_\rho : \gamma \in I\}$ satisfies Proposition 2.2. Its simple real points are precisely the

simple critical zeros. Their real-coordinate span is at most $X_T = HL/(2\pi)$. The short-interval Riemann–von Mangoldt formula gives

$$N(T, H) = X_T + O(H + \log T), \quad X_T/N(T, H) \rightarrow 1. \quad (21)$$

After division by N , the error in (19) is $O_f(1/L + T^{\lambda-\theta}L) = o(1)$. The strict support gap $\lambda < \theta$ is therefore necessary at this stage.

The cosine density (13) is Wang’s minimizing density and satisfies

$$\mathcal{C}(f_\lambda) = \frac{\lambda}{2} + \frac{1}{\sqrt{2}} \cot(\lambda/\sqrt{2}) = 2 - c(\lambda). \quad (22)$$

It is not a smooth compactly supported test because its endpoint values do not vanish. Choose even cutoffs $0 \leq \chi_\varepsilon \leq 1$, supported strictly inside $(-\lambda/2, \lambda/2)$ and approaching one in the interior. Normalizing $\eta_\varepsilon = \chi_\varepsilon \sqrt{f_\lambda} / \|\chi_\varepsilon \sqrt{f_\lambda}\|_2$ gives densities $f_\varepsilon = \eta_\varepsilon^2$ converging in both L^1 and L^2 to f_λ . Moreover $f_\lambda \rightarrow f_\theta$ in those spaces as $\lambda \uparrow \theta$, and $\mathcal{C}$ is continuous for these convergences on bounded support [3, Section 4].

For normalized nonnegative densities,

$$\sup_{x \in \mathbb{R}} |\widehat{f}(x) - K_\theta(x)| \leq \|f - f_\theta\|_1, \quad |\widehat{f}(x)|, |K_\theta(x)| \leq 1. \quad (23)$$

Thus the doubled triple energy changes by at most $12\|f - f_\theta\|_1$. This retains a fixed positive energy margin through the smooth approximation.

Proposition 5.1 (Transfer with a certified limiting gap). *Fix $0 < \theta < 1$, $R > 0$, and $0 < d \leq 1$. Suppose (11) holds for K_θ . Define*

$$c_d = \frac{c(\theta) - 2d/(3R)}{1 - d/3}.$$

Then the simple-critical and distinct proportions in (5) are at least c_d and $(1 + c_d)/2$, respectively.

Proof. First fix $0 < d' < d$. Choose a sequence of fixed smooth densities f_j supported in $(-\lambda_j/2, \lambda_j/2)$, with $\lambda_j < \theta$, converging to f_θ in L^1 and L^2 . By (23), every sufficiently large j has triple energy at least d' . Fix such a j before letting T grow.

Write $s = N_0^s(T, H)$, $N = N(T, H)$, and Q for the pair sum with this fixed kernel. Propositions 2.2 and Lemma 3.2, with actual span at most X_T , imply

$$\left(1 - \frac{d'}{3}\right) \frac{s}{N} \geq 2 - \frac{Q}{N} - \frac{2d'}{3N} - \frac{2d'X_T}{3RN}.$$

Let $T \rightarrow \infty$, using (19) and (21), and then let $j \rightarrow \infty$. The result is $\liminf s/N \geq c_{d'}$.

For the distinct count, the other finite inequality gives

$$\liminf \frac{N^d(T, H)}{N} \geq \frac{1 + c(\theta)}{2} + \frac{d'}{6}(c_{d'} - 2/R) = \frac{1 + c_{d'}}{2}.$$

The equality uses $c_{d'} - c(\theta) = d'(c_{d'} - 2/R)/3$. Finally let $d' \uparrow d$; all scalar expressions are continuous since $d \leq 1 < 3$. No test function changes inside a large- T limit. $\square$

Proof of Theorem 1.1. Lemma 4.1 with $\lambda = \theta$ proves a limiting-kernel gap at least $2(b/B)^2$, and hence at least the positive δ in (3). Apply Proposition 5.1 with $d = \delta$. Its constant equals (4). The gain is strictly positive because $R > 2/c(\theta)$. $\square$

6 The certified numerical example

For $\theta = 3/4$ and $R = 21/4$, the recorded finite computation proves

$$2\{K_{3/4}(u)^2 + K_{3/4}(v)^2 + K_{3/4}(u+v)^2\} \geq \frac{1}{7000} \quad (u, v \geq 0, u+v \leq 21/4). \quad (24)$$

The certificate is a complete binary partition of the square $[0, R]^2$. Every endpoint and threshold is rational. A box is either subdivided, proved wholly outside the triangle, or accepted using an energy lower bound valid on the entire box. The final tree contains 48,761 nodes, 24,252 accepted energy leaves, 129 outside leaves, and no unresolved boxes [8].

For completeness, the analytic enclosure used at each leaf is simple. Since f_θ is a probability density on $[-\theta/2, \theta/2]$,

$$|K'_\theta(x)| \leq 2\pi \int |u|f_\theta(u)du \leq \pi\theta. \quad (25)$$

On a real interval with center x_0 and radius r , an outward interval evaluation of $K_\theta(x_0)$ therefore supplies the valid lower bound

$$|K_\theta(x)| \geq \max\{0, \text{lower}(|K_\theta(x_0)|) - \pi\theta r\}.$$

Apply this separately to the box ranges for u , v , and $u+v$, square the nonnegative lower bounds, and double their sum. Every acceptance comparison requires the entire lower enclosure to exceed the rational threshold. An outside box is discarded only if the sum of its two lower coordinates is strictly greater than R , so the triangle boundary is retained.

Discovery used Arb through `python-flint==0.9.0` at 160-bit precision and the removable-singularity-free formula (15). Each accepted leaf was replayed at 256 bits with a separately derived Taylor evaluator. For real x ,

$$\text{sinc}(x) = \sum_{j=0}^{95} \frac{(-1)^j x^{2j}}{(2j+1)!} + \mathcal{R}(x), \quad |\mathcal{R}(x)| \leq \frac{|x|^{192}}{193!}. \quad (26)$$

The remainder follows by applying the real sine Taylor theorem through degree 192 and dividing by x , with continuous extension at zero. Arb encloses both the polynomial and its remainder. This replay is an independent kernel evaluator; it shares the partition reconstruction, Lipschitz argument, and Arb library with the first computation. It is not a second independent full verifier or a Lean proof.

Proof of Corollary 1.2. Use the finite certificate (24) in Proposition 5.1, with $d = 1/7000$ and $R = 21/4$. The full threshold d is available because the proposition first uses every $d' < d$ through smoothing and then passes to d . Substitution gives the stated constants. The outward rational enclosures for their values and positive difference are recorded with the certificate. $\square$

The certificate's role is quantitative. Even without it, Theorem 1.1 proves a strict improvement for every fixed $\theta > \theta_0$ using its explicit analytic gap. Conversely, the finite numerical inequality alone does not prove a theorem about zeta zeros without the finite operator and fixed-test analytic transfer.

Proportion at $\theta = 3/4$	Wang baseline	This certificate
Simple and critical	0.4190750129754243...	0.4190768284253039...
Distinct	0.7095375064877121...	0.7095384142126519...

Table 1: All denominators count nontrivial zeros with multiplicity in $(T, T + T^{3/4}]$. The entries are proportions, not percentages.

A Reproducibility, provenance, and scope

The public record [8], experiment **EXP-002-short-interval-stability**, contains the mathematical proof, an adversarial audit, the interval runner, the canonical partition tree, and rational enclosures for the constants. The hypothesis and permitted parameter family were declared before computation in repository commit **266486f**. A deterministic floating grid selected candidate radii; its sampled values were never used as certified lower bounds. The final certification used CPU arithmetic; no GPU was necessary.

From the repository root, with the problem’s pinned requirements installed, a fresh output directory preserves the canonical artifacts:

```
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-002-short-interval-stability/run.py \
    --output-dir tmp/riemann-replay --radius 21/4 --threshold 1/7000
pytest tests/test_riemann_certificates.py
```

The displayed backslashes indicate line continuation for readability; use one command line on shells with different continuation syntax. The runner fails with an exact resumable checkpoint if its time or node budget expires while boxes remain. It does not report a certificate in that state. A separate bounded trial exercised termination and exact resume behavior.

The tree replay checks its hash, reconstructs every box, verifies all accepted leaves, and rejects missing, trailing, unknown, or miscounted nodes. A hash protects accidental corruption; geometric replay is still required when a modified tree has a recomputed hash. Adversarial checks rejected false whole-square acceptance, a false outside-domain claim, truncation, appended data, altered counts, an unresolved flag, an invalid exponent, and an inflated threshold. Nineteen focused tests, including one complete 256-bit certificate replay, passed in the recorded run. These are implementation checks, not formal verification of the entire analytic proof.

The canonical preorder tree has SHA-256 digest

78829635026aeacd69c02213cc899b5029ef2566bbfaf1c86b757cc5f92ffe55

The source comparison was completed on 2026-09-12. It used the full Wang v1 manuscript, Lamzouri v2, the current Alpöge–Furman paper, and pinned snapshots of the directly relevant global stability repositories. Queries combining the Wang identifier, short intervals, stability, Gram defects, and exact numerical signatures found no prior public statement of this short-interval refinement. This supports the scoped contribution claimed here, without establishing exhaustive priority or detecting unpublished concurrent work. The known finite stability inequality and cosine-root mechanism are explicitly credited to the repository identifier **ainta**; no personal author identity is inferred from that identifier.

The theorem relies on established classical analytic inputs through Wang’s recent preprint, not on an end-to-end machine formalization of those inputs. The analytic and numerical reviews can

still contain errors, and the two numerical evaluators share arithmetic and geometric components. No effective height threshold, uniform exponent limit, lower positivity threshold, global-record claim, or elimination of all off-line zeros follows from this work.

AI assistance was used for derivation, source research, adversarial review, and implementation. These automated reviews do not constitute independent peer review. The new manuscript and original research exposition are released under CC BY 4.0; code follows the repository’s MIT license. Cited third-party materials retain their own licenses. In particular, local copies of arXiv texts with nonexclusive-distribution licenses are not relicensed by this manuscript.

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